Geometric Distribution

Suppose that there is aseries of independent trails and in each trail the probability of success p remains constant then the probability of first success after X=k number of failures is given by $q^k p$.

A discrete r.v is said to follow geometric distribution if the pmf of the r.v. X is as follows

$prob(X = k) = q^k p$, where $k = 0,1,2,...$ where p+q=1.

In Geometric distribution the r.v. X takes only non negative values.

Assumptions of Geometric Distribution:

- The trails are independent.
- There are only two possible outcomes for each trial, eithre success or failure.
- The probability of success, p, is the same for every trial.

Properties of Geometric distribution:

- It is a single parameter distribution $p = \frac{1}{\mu}$
- Geometric distribution lacks memory, i.e., $P(X \geq s + t \mid X \geq t) = P(X \geq s)$

Proof: A geometric random variable X has the memoryless property if for all nonnegative integers s and t,

$$\text{Prob}\ (X \geq s + t \mid X \geq t) = \text{Prob}\ (X \geq s)$$

The probability mass function for a geometric random variable X is

$$f(x) = q^x p, \ \ x = 0, 1, 2, \ldots.$$

The probability that X is greater than or equal to x is

$$P(X >= x) = q^x, \ \ x = 0, 1, 2, \ldots.$$

So the conditional probability of interest is

$$\text{Prob}(X \geq s + t \mid X \geq t) = \frac{\text{Prob}[(X \geq s + t) \cap \text{Prob}(X \geq t)]}{\text{Prob}(X \geq t)}$$

$$= \frac{\text{Prob}[(X \geq s + t)]}{\text{Prob}(X \geq t)}$$

$$= \frac{q^{s+t}}{q^t} = q^s = prob.(X \geq s)$$

which proves the memory less property.

Moment generating function of Geometric Distribution

$$mgf = M_X(t) = E(e^{tX}) = \sum_{k=0}^{\infty} e^{tk} \times prob(X = k) = \sum_{k=0}^{\infty} e^{tk} \times q^k p = p + e^t qp + e^{2t} q^2 p + e^{3t} q^3 p + ...$$

$$= p[1 + qe^t + (qe^t)^2 + ..] = \frac{p}{(1 - qe^t)}$$

Hence the mean and variance of Geometric distribution are

mean = $E(X) = M_x'(t)\ at\ t = 0 = \frac{q}{p}$ and variance is $\sigma^2 = [M_x''(t) - M_x'(t)]$ at $t = 0 = \frac{q}{p^2}$

Examples

1. You play a game of chance that you can either win or lose (there are no other possibilities) until you lose. Your probability of losing is p=0.57. What is the probability that it takes five games until you lose?

Ans. Prob(X=5) = $(0.43)^5(0.57)$

2. A die is cast until 6 appears. what is the probability that it must be cast more than five times.

Ans. $prob(x > 5) = 1 - prob(X \le 5) = 1 - \sum_{k=1}^{5} (\frac{5}{6})^{k-1} \times (\frac{1}{6})$

3. If your probability of success is 0.2, what is the probability you meet an independent voter on your third try?

Ans. Here p=0.2 and k=3. Prob(X=3) = $(1\text{-}0.2)^2(0.2)$

4. You throw darts at a board until you hit the center area. Your probability of hitting the center area is $p=0.17$. You want to find the probability that it takes eight throws until you hit the center. What values does *X* take on?

5. A safety engineer feels that 35% of all industrial accidents in her plant are caused by failure of employees to follow instructions. She decides to look at the accident reports (selected randomly and replaced in the pile after reading) **until** she finds one that shows an accident caused by failure of employees to follow instructions. On average, how many reports would the safety engineer **expect** to look at until she finds a report showing an accident caused by employee failure to follow instructions? What is the probability that the safety engineer will have to examine at least three reports until she finds a report showing an accident caused by employee failure to follow instructions?